

Converting ER Model to Relational Tables

Q. How is an ER Diagram mapped into a Relational Database schema? Explain the rules for converting Entities, Attributes, and the various Relationships (1:1, 1:N, M:N) into database tables.

> **Definition to Remember**

1. Rule 1: Strong Entities and Attributes

- Every **Strong Entity Type** becomes a separate **Table**.

2. Rule 2: Multi-Valued Attributes

- Cannot store a multi-valued attribute in the main table (relational tables require **atomic** values).

3. Rule 3: Weak Entities

- Weak Entity converted into a **separate table** with its own attributes.

4. Rule 4: Mapping Binary Relationships

| Cardinality | New Table Needed? | Action |

1:N Mapping:

M:N Mapping:

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

> `Str
Table)